



SEMESTER 2 EXAMINATIONS 2018/2019

MODULE: CA267 - Software Testing

PROGRAMME(S):

CASE BSc in Computer Applications (Sft.Eng.)
ECSAO Study Abroad (Engineering & Computing)

YEAR OF STUDY: 2,O

EXAMINER(S):

Dr. Paul Clarke (Internal) (Ext:7021)

TIME ALLOWED: 2 Hours

INSTRUCTIONS: Answer all questions.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.
The use of programmable or text storing calculators is expressly forbidden.

There are no additional requirements for this paper.

QUESTION 1**[TOTAL MARKS: 25]****Q 1(a)****[12 Marks]**

Identify four different types of test team roles and provide a detailed description for each role type.

Q 1(b)**[5 Marks]**

Using an example, briefly describe what is involved in dynamic testing.

Q 1(c)**[4 Marks]**

In your opinion, what is the first stage of dynamic software testing? Justify your opinion with clear reasoning.

Q 1(d)**[4 Marks]**

Provide a description of boundary value analysis, using an example to demonstrate how it is applied in practice.

[End of Question1]**QUESTION 2****[TOTAL MARKS: 25]****Q 2(a)****[10 Marks]**

When writing unit test cases, outline an effective strategy that you would recommend to a programmer? Justify your strategy with a clear explanation.

Q 2(b)**[5 Marks]**

Assuming that π (π) = 22/7, demonstrate how PyUnit would be used to check the π calculation to 2 decimal points of precision. Provide only the line of code required for the test and include the formula for calculating π as a parameter to the assert statement.

Q 2(c)**[10 Marks]**

In your own words, describe how defect severity and priority can be used in combination to determine the order in which defects can be resolved.

[End of Question2]

QUESTION 3**[TOTAL MARKS: 25]****Q 3(a)****[6 Marks]**

Identify and briefly explain the three different types of anomalies identifiable by data flow analysis.

Q 3(b)**[9 Marks]**

In your own words, explain why anomalies identified in data flow analysis are considered important in terms of software quality. Use examples to support your explanation.

Q 3(c)**[10 Marks]**

In the following python code, identify any data flow anomalies that may exist. When identifying the anomaly, highlight the line of code where the anomaly arises, the anomaly type and specific details of the anomaly. (Note that it is not expected to evaluate the code other than to explicitly identify data flow anomalies).

```
1 def is_leap_year():
2
3     year = 0
4     leap_year = False
5     new_year = leap_year
6
7     if (year < 0):
8         return False
9
10    year = 2019
11    if (year % 4) == 0:
12        if (year % 100) == 0:
13            if (year % 400) == 0:
14                print("{0} is a leap year".format(year))
15            else:
16                print("{0} is not a leap year".format(year))
17        else:
18            print("{0} is a leap year".format(year))
19    else:
20        print("{0} is not a leap year".format(year))
21        finished = True
22
23    leap_year = True
24
25    is_leap_year()
```

[End of Question3]

QUESTION 4**[TOTAL MARKS: 25]****Q 4(a)****[7 Marks]**

Explain why software test estimation is referred to as "indeterminate". As part of your response, identify four distinct reasons why estimating software testing effort is indeterminate.

Q 4(b)**[6 Marks]**

In the context of the python IDLE debugger, explain the difference between the "Step" and "Over" commands as available from the IDLE debug window. Clearly explain the rationale for having both debugging commands.

Q 4(c)**[12 Marks]**

Identify a robust method for testing software requirements, providing examples and a detailed explanation.

[End of Question4]***[END OF EXAM]***